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EXPT.NO - 8

TITLE - Dijkstra's Algorithm

```
In [2]: 1 # Taking inputs from the user
2 n = int(input("Enter number of vertices in the graph: "))
3
4 print("Enter the adjacency matrix where each row is separated by spaces:")
5 graph = []
6 for i in range(n):
7     row = input().split()
8     graph.append([int(x) for x in row])
9
10 source = int(input("Enter the source vertex: "))
11
12 def dijkstra(graph, source):
13     V = len(graph) # Number of vertices
14
15     # Initialize distances from source to all vertices as infinity
16     dist = [float('inf')] * V
17     dist[source] = 0
18
19     # Track visited vertices
20     visited = [False] * V
21
22     for _ in range(V):
23         # Find the vertex with minimum distance not yet visited
24         min_dist = float('inf')
25         min_index = -1
26
27         for v in range(V):
28             if not visited[v] and dist[v] < min_dist:
29                 min_dist = dist[v]
30                 min_index = v
31
32         # Mark the selected vertex as visited
33         visited[min_index] = True
34
35         # Update distances of adjacent vertices
36         for v in range(V):
37             if not visited[v] and graph[min_index][v] != 0:
38                 dist[v] = min(dist[v], dist[min_index] + graph[min_index][v])
39
40     return dist
41
42 # Function call
43 shortest_paths = dijkstra(graph, source)
44
45 print("=====")
46 print("Shortest distances from node", source, "to each node:")
47 for i, d in enumerate(shortest_paths):
48     print("Vertex", i, "Distance =", d)
49 print("=====")
```

```
Enter number of vertices in the graph: 5
Enter the adjacency matrix where each row is separated by spaces:
0 0 0 7 0
3 0 4 0 0
0 0 0 5 6
0 2 0 0 0
0 0 0 4 0
Enter the source vertex: 0
=====
Shortest distances from node 0 to each node:
Vertex 0 Distance = 0
Vertex 1 Distance = 9
Vertex 2 Distance = 13
Vertex 3 Distance = 7
Vertex 4 Distance = 19
=====
```